

Initial Project Planning Template

Date	15 July 2026
Team / Submission ID	
Project Name	Emotion Detection & Learning Support Engine
Submitted By	Venu Gopal (Sole Team Member / Team Lead)
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation

As a solo contributor, all sprints were planned, estimated and executed by Venu Gopal. Story points follow a simple 1 (few hours) – 2 (half day) – 3 (1-2 days) scale.

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start	Sprint End (Planned)
Sprint-1	Data Preparation	USN-1	As a developer, I can collect and clean the emotion-labelled text datasets (eng_dataset.csv, emotion-emotion_69k.csv) for training.	2	High	Venu Gopal	01 Jun 2026	05 Jun 2026
Sprint-1	Text Preprocessing	USN-2	As a developer, I can build a preprocessing pipeline (lowercasing, cleaning, tokenising, stop-word removal) in src/preprocessing.py.	1	High	Venu Gopal	01 Jun 2026	05 Jun 2026
Sprint-2	BiLSTM Model	USN-3	As a developer, I can train and save a BiLSTM emotion classifier with its tokenizer and label encoder.	3	High	Venu Gopal	06 Jun 2026	12 Jun 2026
Sprint-2	BERT Model	USN-4	As a developer, I can fine-tune a BERT classifier for emotion detection and export it for inference.	3	High	Venu Gopal	06 Jun 2026	12 Jun 2026
Sprint-3	Mixed Emotion Logic	USN-5	As a developer, I can detect secondary/mixed emotions and boost predictions using keyword rules.	2	Medium	Venu Gopal	13 Jun 2026	18 Jun 2026
Sprint-3	Gemini Integration	USN-6	As a learner, I can receive an AI-generated, empathetic response based on my detected emotion and problem.	2	High	Venu Gopal	13 Jun 2026	18 Jun 2026
Sprint-3	Fallback Responses	USN-7	As a learner, I still receive a helpful response even when the Gemini API is unavailable.	1	High	Venu Gopal	13 Jun 2026	18 Jun 2026

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start	Sprint End (Planned)
Sprint-4	Streamlit UI	USN-8	As a learner, I can select my subject, describe my problem, and view model predictions on one page.	2	High	Venu Gopal	19 Jun 2026	25 Jun 2026
Sprint-4	Logging	USN-9	As a developer, every interaction is saved to a CSV file with a timestamp for later analysis.	1	Medium	Venu Gopal	19 Jun 2026	25 Jun 2026
Sprint-5	Analytics Dashboard	USN-10	As a learner, I can view charts of my emotion distribution, emotional journey, and field-wise trends.	2	Medium	Venu Gopal	26 Jun 2026	02 Jul 2026
Sprint-5	Testing & Polish	USN-11	As a developer, I can test predictions end-to-end and fix issues found during manual testing.	2	Medium	Venu Gopal	26 Jun 2026	02 Jul 2026
Sprint-6	Documentation & Submission	USN-12	As a developer, I can prepare full project documentation and package the repository for submission.	2	High	Venu Gopal	03 Jul 2026	15 Jul 2026